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Unlocking Transparency in Credit Scoring: Leveraging XGBoost with XAI for Informed Business Decision-Making

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Abstract*—Credit score analysis is vital to modern banking systems, allowing banks and other financial institutions to determine a borrower's creditworthiness. In such a situation, accurate and robust prediction models are vital because they allow lenders to make rational decisions regarding loan approvals and risk management. This paper provides an overview of using XGBoost, a sophisticated machine learning algorithm, to improve credit score evaluation, and the XAI model, LIME, to describe the black box machine learning algorithm. XGBoost, a gradient boosting-based ensemble learning algorithm, has gained prominence for its capacity to give improved predicted accuracy while handling vast and complicated datasets. Its algorithmic characteristics, including regularization, parallel processing, and decision tree optimisation, make it especially well-suited for credit scoring problems. Because of its complexity, implementing XAI is critical since it will help lenders grasp the reasons for the result of the XGBoost. The results show how the XAI model, LIME, helps simplify the complexity of these models. It is critical to integrate XAI models since they will improve lender decision-making. The fundamental goal of this research is to evaluate the XAI model, LIME, and determine how well the XAI model explains the findings of our experimental tests. Furthermore, it illustrates the possibility of incorporating LIME into credit score analysis, resulting in more efficient lending procedures, enhanced risk management, and better decision-making. Finally, this paper emphasizes the importance of using advanced machine learning techniques such as XGBoost in credit scoring analysis, which has the potential to transform the way banks and other financial institutions assess credit risk, as well as include LIME for a better understanding of the results.

***Keywords*—credit scoring, XGBoost, XAI, LIME, Blackbox, algorithm.**[1]

I. Introduction

Credit scoring has been around but has become progressively more complicated in recent years. Lenders use it as a statistical approach to assess a borrower's creditworthiness. It also estimates a borrower's likelihood of falling short on a loan. Credit scores are determined by several factors, including the borrower's credit history, income, and debt-to-income ratio. It may also be a projection of credit behaviour based on data acquired from credit history, such as the likelihood of repaying a loan on time. One of the fundamental problems with credit scoring is that it depends on unclear conditions, with the danger of borrowers failing to meet their obligations [1]. Furthermore, in credit score studies, the use of deep learning algorithms for classification has primarily gone unnoticed [2]. As a result, it is a problematic decision-making challenge that may compromise the survival of an organization. Using deep learning algorithms for categorisation is crucial in credit scoring since it reduces the challenges of decision-making tasks. Moreover, this study aims to introduce the use of XGBoost, a sophisticated machine learning algorithm for improving credit score evaluation, and the XAI model, LIME, to explain the black-box nature of machine learning algorithms. Furthermore, the objective of this study is:

1. To evaluate the effectiveness of LIME in explaining the results of experimental tests in credit scoring.
2. To assess the impact of LIME on the decision-making process in credit scoring.
3. To identify potential challenges and limitations of using LIME in credit scoring.

As previously mentioned, the primary goal of this research is to examine the XAI model, LIME, and how well the XAI

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model explains the results of our experimental tests using XGBoost as our classifying algorithm.

II. Related Literature

Explainable artificial intelligence (XAI) is an area of artificial intelligence study that attempts to provide ways to make machine learning models more interpretable. The ability to comprehend models is vital because it helps us understand how they function, which attributes are most relevant to them, and why they make their predictions. It is particularly significant in high-stakes fields like healthcare, banking, and criminal justice, where trust in AI systems' conclusions is critical. One of the difficulties with understanding is that machine learning models, intense learning models, can be quite complicated. Models like these may learn complicated correlations between characteristics and outcomes, but understanding how they do so can be challenging. In [3], the study provides a mechanism for producing comprehensible explanations of credit scoring models. Furthermore, to fulfil the demands of diverse users, the framework employs a range of strategies, including global explanations, local feature-based explanations, and local instance-based explanations [3]. Moreover, global explanations offer a broad model overview, such as which elements are essential. Additionally, Local feature-based explanations describe how a particular attribute influenced the model's prediction for a specific occurrence. Local instance-based explanations describe how the model's prediction might vary if a specific instance was changed. Furthermore, the work in [3] is classified using an Extreme Gradient Boosting (XGBoost) model. The researchers picked XGBoost since it is a strong and versatile classification model that is also reasonably easy to comprehend. This implies that it is feasible to comprehend how the algorithm makes its predictions, which is critical in high-stakes areas like credit scoring. Aside from XGBoost, the research in [3] looked at a variety of additional classification models, such as logistic regression (LR), decision trees (DT), random forests (RF), and artificial neural networks (ANN). XGBoost, on the other hand, surpassed all the other models in terms of precision and clarity. Furthermore, [4] proposed a strategy for employing explainable artificial intelligence (XAI) to increase the comprehensibility of credit scoring models. The strategy depends on an approach known as SHAP values, which measures every feature's impact on the prediction made by the model [4]. The approach is tested on a data set of unprotected consumer loans obtained from a Norwegian bank [4]. Furthermore, it has been demonstrated that the technique can produce precise and understandable explanations for the model's predictions. Another discovery is that the explanations can be utilized to detect any prejudices in the model and improve its fairness [4]. The classification model employed in the paper "Explainable AI for Credit Assessment in Banks" was the Light Gradient Boosting Machine (LightGBM). LightGBM is a well-known gradient-boosting method with excellent precision and efficiency. It is also straightforward to comprehend, making it an excellent candidate for XAI applications. In addition, the LightGBM model was integrated with the SHAP (Shapley Additive exPlanations) approach to create comprehensible interpretations for the model's predictions [4]. SHAP values are a comprehension approach describing how each unique feature contributes to the model's prediction. Furthermore, [5] study assesses the capability of two ordinary explainable artificial intelligence (XAI) models, SHAP and LIME, to differentiate among data groups. XGBoost was adopted [5]. Additionally, [6] stressed out the fundamental issue is that algorithms are considered "black box models" that cannot meet legal requirements. Although deep learning approaches like LSTM have been examined for behavioural credit scoring based on performance, the study has yet to assess these algorithms comprehensively based on efficiency and determining explainability. Additionally, [6] has only used one XAI model, SHAP, for global and local explanation. In addition, although no model generates significant outcomes, XGBoost is the greatest classifier in accuracy. Although all studies concur that XAI is essential for fostering trust in classifier algorithms, there are significant differences in their theoretical frameworks and procedures. Conventional data mining workflows are widely used for credit risk assessment and anomaly identification. Interpretable neural networks and attention mechanisms further enhance the methodological tapestry. This diversity displays a dynamic environment where academics actively investigate different strategies to achieve successful and interpretable classification, reflecting the field's developing character. Furthermore, this study employs many datasets with a complicated classifier to examine how well XAI explains biased and unbalanced datasets.

III. Research Methodology

A. *Workflow*

1. Data Preprocessing: Data preparation, which includes data cleansing, feature selection, and imputation, is an essential stage in credit score analysis. The resilience of XGBoost to missing data and its ability to handle both categorical and continuous variables make it a powerful tool for this stage. However, data imputation will not be implemented in this study.
2. Model Training: The gradient-boosting architecture used by XGBoost enables the algorithm to learn complicated patterns and correlations in credit data. Its capacity to combine numerous weak models into strong ones results in a credit-scoring model that is very accurate.
3. Model Evaluation: Different performance indicators, such as accuracy, confusion matrix, and the Area under the curve (AUC), will be used to quantify the predictive capacity of the XGBoost credit scoring model.
4. LIME: Despite the complex structure of XGBoost, the XAI model, LIME, will simplify the outcomes for easier comprehension.

B. *Classification technique(XGBoost)*

XGBoost is a tree ensemble model that addresses the limits of individual trees by aggregating their predictions through a linear combination of many trees. The ensemble model exhibits a progressive enhancement in its predictive capabilities through an iterative error learning process and novel data assimilation. Moreover, XGBoost is characterized

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by its regularization technique, which enables the manipulation of variable weights, mitigating the issue of overfitting and improving variable selection in situations involving many dimensions. The ultimate model is a fusion of all the trees inside the ensemble [7]. The method assesses each leaf and, if deemed required, generates additional partitions to analyze the weak learner's impact on the model. If this contribution is essential, the algorithm constructs new trees sequentially. During this process, the algorithm integrates the values from the preceding tree as new predictions while simultaneously diminishing the residuals of the new trees by applying a learning rate. The model's performance is evaluated during each iteration using a loss function that compares the observed and predicted data. This loss function may be customized to align with a particular continuous probability distribution [8].

C. *Explainable AI Techniques(LIME and SHAP)*

The Local Interpretable Model-Agnostic Explanation (LIME) framework is an open-source tool that aims to increase confidence in machine learning models by explaining their decision-making processes. The analysis focuses on particular data, with the objective of rendering models interpretable and independent of any given model. LIME offers valuable insights into the inner workings of models, including the ability to highlight certain picture areas or pinpoint significant aspects. Its primary capabilities encompass image interpretation, text analysis, and tabular data assessment [9]. The process of LIME involves many sequential phases. Firstly, the original model is trained. Following this, variables are carefully chosen to provide explanations. Subsequently, neighbouring data points are generated using perturbation. These points are then assigned weights. A simple model is then fitted using these weighted points. Finally, the behaviour of the complicated model at a given data point is elucidated using the fitted simple model [10]. Nevertheless, how different is LIME from SHAP? The Shapley Additive explanations (SHAP) methodology was first introduced by Shapley in 1953, drawing upon principles from the field of game theory. The approach above provides a robust and wise assessment of the significance of a particular attribute inside a given model [11]. The SHAP framework is a comprehensive approach to evaluating machine learning models, wherein each feature is assigned a relevance value for a specific prediction. Moreover, the underlying reasoning behind the SHAP methodology is based on the notion that for intricate models such as ensemble techniques or deep networks, a more straightforward explanatory model exists that can be specified as an accessible estimate of the initial structure [12]. According to existing research, incorporating LIME and SHAP into machine learning processes has proven beneficial in augmenting accident detection models by offering further explanations [13]. However, we will be implementing LIME for this study.

IV. Experimental setup

A. *Datasets*

Three datasets were used for our experimental setup and work in this study on one single classifier, XGBoost. The datasets can be accessed publicly from Kaggle. Additionally, the following table contains a list of the datasets:

TABLE I. DESCRIPTION OF THE DATASETS USED IN THE STUDY

Datasets	#Loans	#Good loans	#Bad Loans
Taiwanese	30,000	6,636	23,364
Australian	690	307	383
Credit Risk	32,581	7,108	25,473

The Taiwanese dataset consists of 30,000 applicants, with 6,636 customers being granted a loan and 23,364 customers being denied a loan. This dataset has been widely used for credit risk training by different researchers. Moreover, the Australian dataset comprises 690 applicants with 307 good loans and 383 bad loans. The Australian dataset has also been widely used for credit risk training. Lastly, the Credit Risk dataset comprises 32,581 applicants with only 7,108 good loans and over 25,000 bad loans.

B. *Performance measures*

The traditional performance measures in the Area of credit scoring embraced the evaluation measurements of this study. These measurements are average accuracy, Type I and Type II errors, Area Under the Curve (AUC) and F1-score. A combination of these measurements, rather than a single measure, is used to measure the predictive performance of the proposed credit scoring model.

From the confusion matrix table, the following calculations are defined:

TABLE II. CONFUSION MATRIX FOR CREDIT SCORING

	Reference Class	
Predicted Class	Good Loans	Bad Loans
Good Loans	TP	FP (Type I error)
Bad Loans	FN (Type II error)	TP

Average Accuracy (ACC) =TP+TN/TP+FN+TN+FP (1)

Type I error = FP/ TN+FP (2)

Type II error = FN/ TP+FN (3)

A TP stands for good applicant, correctly classified as good, TN stands for bad applicant, correctly classified as bad, FN (Type II) stands for the good applicant, incorrectly classified as bad customer, and FP (Type I) stands for bad customer, incorrectly classified as good customer (high risk).

V. Experimental work

This study's experiment used R studio version 4.2.3 equipped with an Intel CORE i5 processor clocked at 1.80 GHz and 8 GB of RAM, running Microsoft Windows 11. The dataset was divided into two sets: a training set used to train the model and a testing set used to assess the constructed XGBoost model. The data distribution was 30% for the testing and 70% for the training set. Furthermore, our final success classifier differs from what this study seeks to

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quantify. As a result, the datasets have no feature selection and have remained unbalanced. This method is used to test how well XAI can explain imbalanced datasets.

VI. Result and Discussion

The usual performance metrics in credit scoring were adopted as the evaluation metrics for this study. These metrics include Type I and Type II errors, Area Under the Curve, ROC Curve, average accuracy, and F1-score. Below is the table description of the predicted class on all datasets' accuracy, AUC, and F1-Score:

TABLE III. DESCRIPTION OF THE PREDICTED ACCURACY AND AUC OF ALL DATASETS

Datasets	Accuracy	AUC	F1-score
Taiwanese	0.820	0.769	0.475
Australian	0.870	0.922	0.858
Credit Risk	0.927	0.940	0.814

The AUC curve depicts a binary classification model's efficiency. It demonstrates the trade-off between the actual positive rate (sensitivity) and the false positive rate (1 - specificity) based on different thresholds for categorizing cases defined by the model. Below is the ROC of all datasets. The AUC results of the three datasets are shown below:

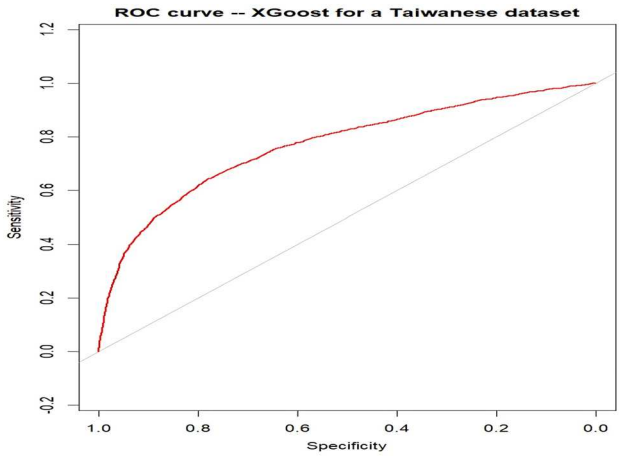
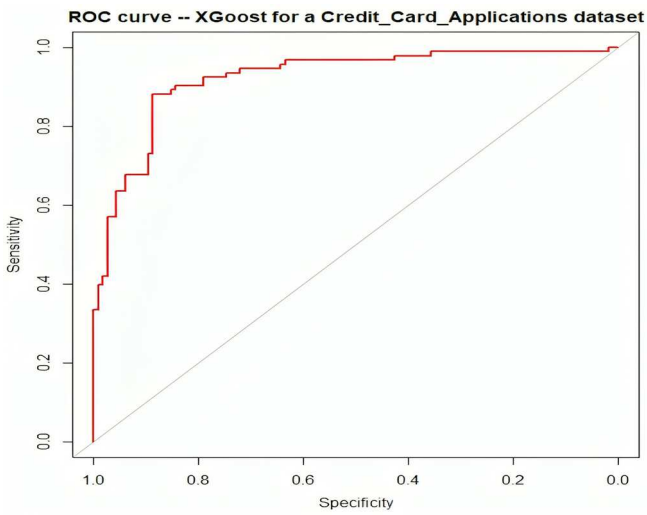


Fig. 1. ROC for TAIWANESE Dataset



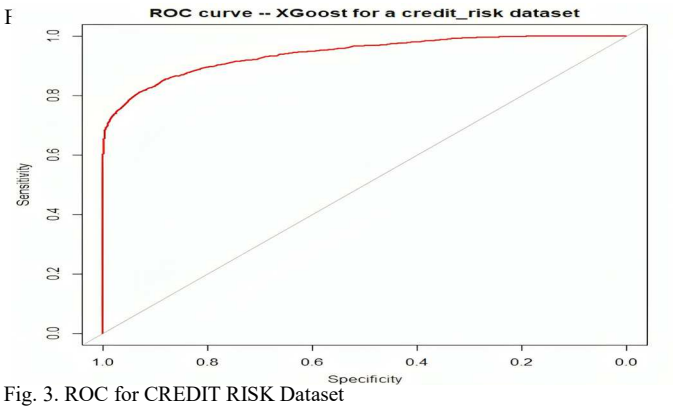


Fig. 3. ROC for CREDIT RISK Dataset

The confusion matrix is a table that displays how many of a categorisation model's predictions were accurate and inaccurate. It is a helpful tool for assessing a model's performance since it might highlight regions in which the model is having difficulty. Below are the figures displaying the confusion matrix of all datasets:

TABLE IV. CONFUSION MATRIX OF TAIWANESE DATASET

	Reference 1	Reference 0
Prediction 1	731	417
Prediction 0	1195	6657

TABLE V. CONFUSION MATRIX OF AUSTRALIAN DATASET

	Reference 1	Reference 0
Prediction 1	82	16
Prediction 0	11	99

TABLE VI. CONFUSION MATRIX OF CREDIT RISK DATASET

	Reference 1	Reference 0
Prediction 1	1566	114
Prediction 0	598	7497

The testing set, or unseen data, provided the basis for all conclusions presented in this section. The Taiwan dataset comprises 6,636 good loans and 23,364 bad loans, the Australian dataset contains 307 good loans and 383 bad loans, and the Credit Risk dataset contains 7,108 good loans and 25,473 bad loans. As was already established, the Taiwanese data's XGBoost accuracy was 82%, which indicates that about 18% of the classifications were incorrectly predicted. Based on an analysis of the confusion matrix, the Taiwanese dataset had 1195 False Negatives (FN) and 417 False Positives. This demonstrates that, despite being good customers, 1195 customers were denied a loan. In addition, FP of 417, the problematic clients, received a loan. Furthermore, the dataset from Australia had an accuracy of 87%, with 13% of the classifications being incorrectly predicted. This dataset's FP was 16, and its FN was 11. This demonstrates that while 16 problematic clients received loans, 11 decent customers had their applications denied. The credit risk dataset had the highest accuracy score in this analysis, at approximately 93%. Nevertheless, nearly 7% of the classification was incorrectly predicted. This dataset's FP was 114, and its FN was 598. This demonstrates that 598 good consumers were turned down for a loan while 114 bad customers received one. In addition, we have included a

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LIME summary to clarify why XGBoost mispredicted these clients considering this issue. Subsequently, we will investigate the Taiwanese dataset's LIME for FP and FN outcomes.

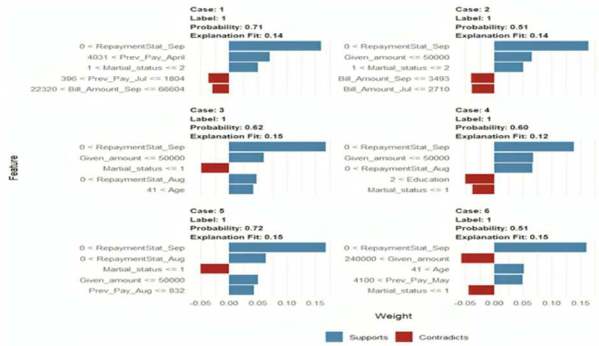

Fig. 4. LIME OF TAIWANESE FP

Fig. 4 illustrates the rationale behind XGBoost's prediction that those unsatisfactory clients would receive a loan (FP). In addition, we have chosen five characteristics at random for lime to explain this result. The red graph shows the effects contradicting the prediction, and the blue graph shows the evidence supporting the conclusion of good customers. For example, it demonstrates that the September repayment status (blue graph) significantly influences the anticipated result of good customers. Additionally, the model's prediction is affected and predicts good clients when the payback status is greater than 0. Moreover, this LIME explanation holds for all FP research outcomes with five randomly chosen attributes.

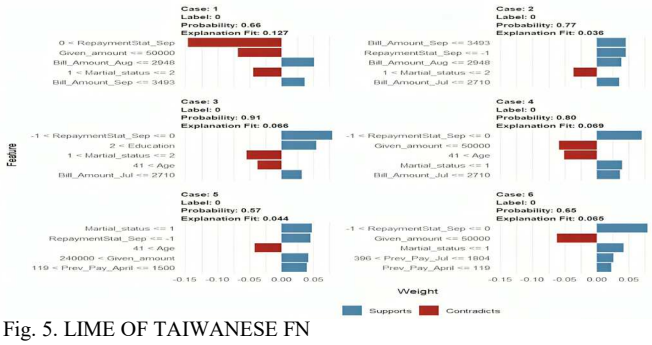

Fig. 5. LIME OF TAIWANESE FN

Fig. 5. demonstrates why the XGBoost model rejected loans to good customers. The red graph has had a more significant impact on the model's outcome. As shown in Fig. 5., when September's repayment status is lower than 0, it contradicts the good customers. Moreover, when the given amount to a customer is equal to or below 50,000, it predicts that the customer will not repay the loan. Hence, in this case, the suitable customer was denied a loan. Moreover, Fig. 6., 7, 8, and 9 demonstrate LIME's explanation to the other datasets.

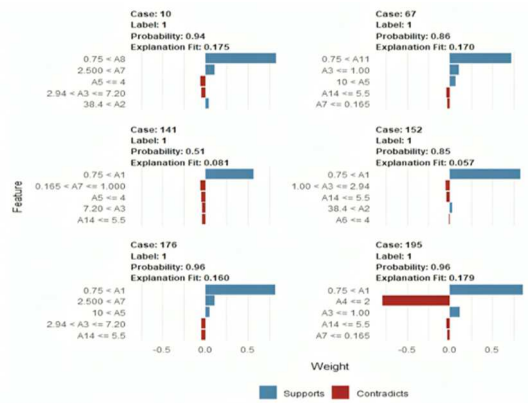

Fig. 6. LIME OF AUSTRALIAN FP

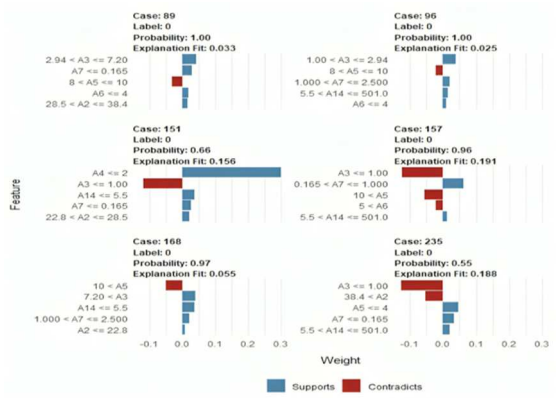

Fig. 7. LIME OF AUSTRALIAN FN

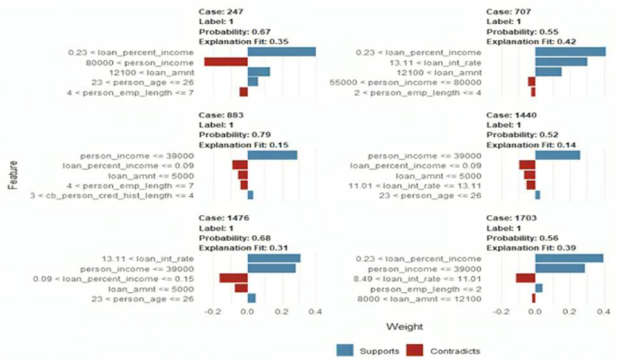

Fig. 8. LIME OF CREDIT RISK FP

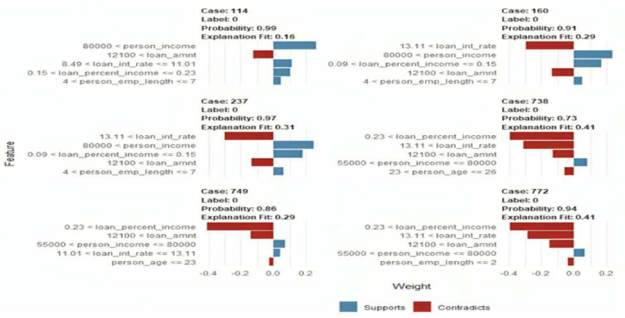

Fig. 9. LIME OF CREDIT RISK FN

Fig. 6. explains why the model offered loans to bad customers in the Australian dataset. In case 10, where attribute 8 is greater than 0.75, the model provides this customer with a loan. Furthermore, in Fig. 7. case 235, when attribute 3 is less than or equal to 1, the model rejects the customer, even if they are good customers. Furthermore, Fig. 8. Case 247 of the credit risk dataset shows that when loan per cent income is higher than 23%, a client is allowed a loan even if they were a bad customer. Finally, Fig. 9. Case 160 shows that if the loan interest rate exceeds 13.11, the model will most likely reject the customer.

VII. Conclusion and Future Direction

The main objective of this study was to examine the XAI model, LIME, and how well it describes the outcomes of our three datasets by utilizing XGBoost as our classifier model.

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The practical work began with collecting datasets (Taiwanese, Australian, and Credit Risk). We subsequently employed the splitting method, dividing the data into 70% training and 30% testing. Finally, the model was trained using the XGBoost classifier. The three datasets average accuracy, type I and type II errors, AUC, ROC curves and F1-score were used to assess the performance of the XGBoost model. Every performance had encouraging outcomes with few mistakes. Afterwards, we used LIME to explain the final results of our XGBoost model. When working with credit score datasets, it is crucial to incorporate XAI models because they can be used to describe any model, regardless of internal workings or complexity. Furthermore, LIME is an effective tool to enhance the interpretability of machine learning models. It is helpful to any machine learning or data science practitioner's toolkit. Moreover, it improves organizational decision-making and task management. Potential directions for future research include.

1. utilizing LIME on heterogeneous ensemble techniques.
2. applying models with greater complexity than XGBoost, such as Deep Neural Networks (DNNs) and
3. Combining LIME and SHAP on various black box classifier types to obtain complete outcomes.

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